A Deterministic Metapopulation SIS Model for Bovine Tuberculosis

Incorporating Heterogeneous Outbreak Phenotypes, Wildlife Reservoir Dynamics, and Cattle-Movement Networks

Mathematical Foundations for Stochastic Simulation

Model Overview and Epidemiological Rationale

We consider a population of cattle herds partitioned into three epidemiological clusters, indexed by $j \in \{1, 2, 3\}$, that were identified through unsupervised clustering of surveillance data. Each cluster represents a distinct outbreak phenotype:

- **Cluster 1** — low transmission, short breakdowns, low recurrence;
- **Cluster 2** — high recurrence despite limited within-herd spread, consistent with persistent wildlife or trade-driven reintroduction;
- **Cluster 3** — intense, sustained outbreaks with high within-herd transmission.

We adopt a *susceptible–infected–susceptible* (SIS) framework because cattle do not acquire lasting immunity following clearance of bovine tuberculosis (*M. bovis*) infection; confirmed reactors are removed under statutory test-and-slaughter, and replacement animals are fully susceptible. The SIS assumption therefore reflects both biological reality and the policy-driven dynamics observed in each cluster.

Each cluster j interacts with the others through (i) the spatial spread of infection across neighboring herds and (ii) the movement of animals between herds. A shared wildlife reservoir $W(t)$ provides an external source of infection and feeds back from cumulative herd prevalence.

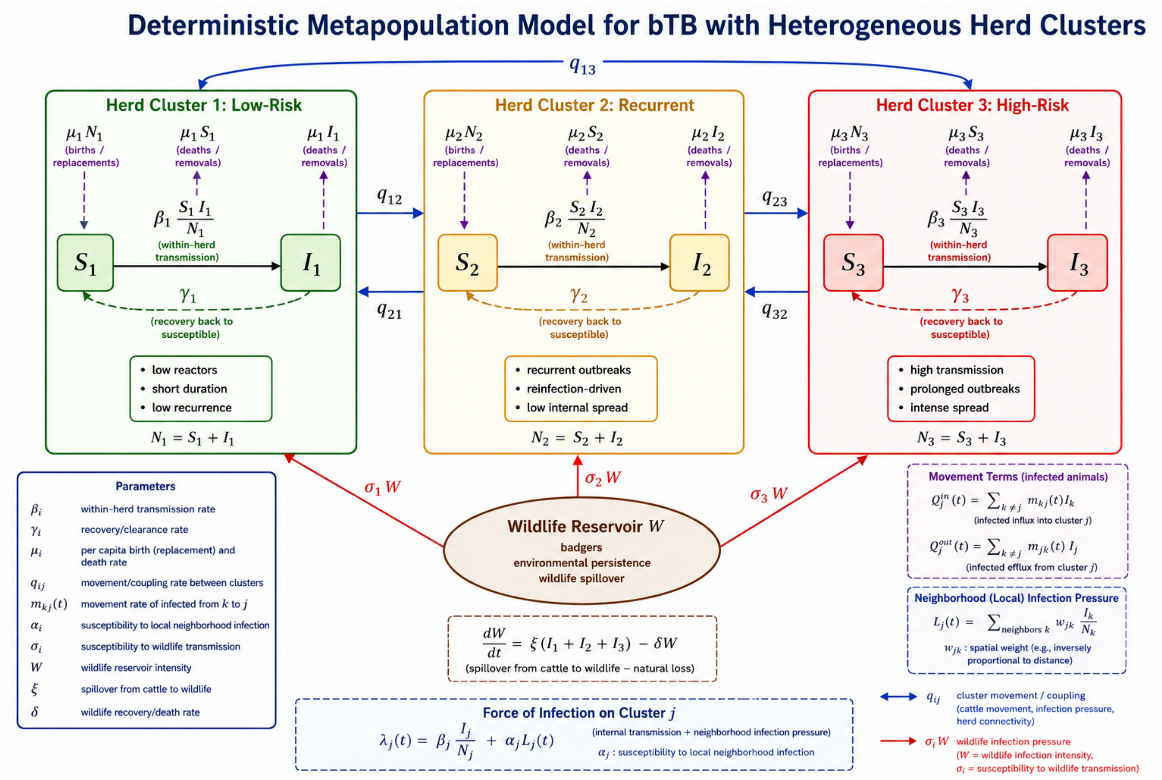

Figure 1. Compartmental diagram of the deterministic metapopulation SIS model for bovine tuberculosis across three heterogeneous herd clusters. Within each cluster j, susceptible (S_j) and infected (I_j) cattle exchange via within-herd transmission (β_j), recovery (γ_j), and demographic turnover (μ_j). Blue bidirectional arrows (q_{jk}) denote inter-cluster animal movements; red arrows ($\sigma_j W$) represent wildlife-to-cattle spillover from the shared badger reservoir $W(t)$, which is itself replenished by infected cattle at rate ξ and depleted at rate δ. The neighbourhood infection pressure $L_j(t)$ and force of infection $\lambda_j(t)$ are defined in the inset boxes and formalised in equations (2) and (3).

State Variables and Notation

Definition 1 (Compartment sizes). *For cluster $j \in \{1, 2, 3\}$ and time $t \geq 0$, let $S_j(t)$ and $I_j(t)$ denote the numbers of susceptible and infectious animals, respectively. The total herd population in cluster j is*

$$N_j(t) \;=\; S_j(t) + I_j(t). \tag{1}$$

The wildlife reservoir is described by a single scalar $W(t) \geq 0$ representing the effective infectious pressure attributable to the badger population and environmental persistence.

Table 1 summarises all parameters, their biological interpretation, and the cluster-specific ordering expected from the surveillance data.

Table 1. Model parameters, biological interpretation, and expected cluster ordering. Inequalities reflect the empirical clustering results; exact values are estimated from surveillance records (see Section).

Parameter	Units	Biological interpretation	Cluster ordering
β_j	day^{-1}	Within-herd transmission rate for cluster j	$\beta_3 \gg \beta_1 > \beta_2$
γ_j	day^{-1}	Rate of recovery / reactor removal returning animals to susceptibility	$\gamma_1 \geq \gamma_2 \approx \gamma_3$
μ_j	day^{-1}	Per-capita natural mortality and replacement rate	Cluster-specific
α_j	dimensionless	Susceptibility to neighbourhood infection pressure	α_2, $\alpha_3 > \alpha_1$
σ_j	day^{-1}	Rate of spillover acquisition from the wildlife reservoir	$\sigma_2 > \sigma_3 > \sigma_1$
w_{jk}	dimensionless	Spatial weight between clusters j and k (inversely proportional to inter-herd distance)	—
$m_{jk}^C(t)$	day^{-1}	Per-capita movement rate of compartment $C \in \{S, I\}$ from cluster j to cluster k	—
ξ	day^{-1}	Spillover rate from infected cattle to the badger/reservoir population	Shared
δ	day^{-1}	Badger population recovery or mortality rate	Shared

Force of Infection

The instantaneous *force of infection* acting on susceptible animals in cluster j decomposes into three epidemiologically distinct pathways.

3.1 Neighbourhood infection pressure. Let $L_j(t)$ denote the weighted prevalence in clusters neighbouring j:

$$L_j(t) \;=\; \sum_{k \neq j} w_{jk} \, \frac{I_k(t)}{N_k(t)}. \tag{2}$$

The weights w_{jk} are inversely proportional to the geodesic distance between the centroid of herds in cluster j and those in cluster k, and are normalised so that $\sum_{k \neq j} w_{jk} = 1$.

This term captures the spatial contagion identified by the multinomial logistic regression, in which proximity to a New Breakdown herd was the strongest predictor of cluster membership.

3.2 Combined force of infection. Aggregating within-herd transmission, neighbourhood spillover, and wildlife-to-cattle spillback, the total force of infection on cluster j is:

$$\lambda_j(t) \;=\; \underbrace{\beta_j \frac{I_j(t)}{N_j(t)}}_{\text{within-herd}} \;+\; \underbrace{\alpha_j\, L_j(t)}_{\text{neighbourhood}} \;+\; \underbrace{\sigma_j\, W(t)}_{\text{wildlife reservoir}} \;. \tag{3}$$

Remark 1. *The three additive terms in equation (3) reflect qualitatively different infection routes: density-dependent transmission within a herd, frequency-dependent spatial contagion from neighbouring herds, and a constant-rate environmental/wildlife source. Cluster 2 is distinguished precisely by the dominance of the third term: σ_2 is elevated relative to β_2, producing frequent breakdowns driven by external reintroduction rather than active within-herd spread.*

Animal Movement Fluxes

Cattle movements between herds are a principal mechanism of long-range disease spread. For each compartment $C \in \{S, I\}$ and each ordered pair (j, k) with $j \neq k$, define:

$$Q_j^{C,\,\mathrm{in}}(t) \;=\; \sum_{k \neq j} m_{kj}^{C}(t)\, C_k(t), \tag{4}$$

$$Q_j^{C,\,\mathrm{out}}(t) \;=\; \sum_{k \neq j} m_{jk}^{C}(t)\, C_j(t), \tag{5}$$

where $m_{jk}^{C}(t)$ is the per-capita rate at which animals in compartment C move from cluster j to cluster k at time t. In practice, $m_{jk}^{I}(t)$ is estimated from cattle-movement databases stratified by source-herd breakdown status, and is expected to be nonzero primarily when the source herd is classified as a New Breakdown (Cluster 2 or 3).

Governing Differential Equations

Cattle population dynamics

For each cluster $j \in \{1, 2, 3\}$, the coupled ordinary differential equations governing $S_j(t)$ and $I_j(t)$ are:

$$\frac{\mathrm{d}S_j(t)}{\mathrm{d}t} = \underbrace{\mu_j\, N_j(t)}_{\text{recruitment}} - \underbrace{\lambda_j(t)\, S_j(t)}_{\text{new infections}} + \underbrace{\gamma_j\, I_j(t)}_{\text{recovery}} - \underbrace{\mu_j\, S_j(t)}_{\text{mortality}} + \underbrace{Q_j^{S,\,\mathrm{in}}(t) - Q_j^{S,\,\mathrm{out}}(t)}_{\text{movement}}, \tag{6}$$

$$\frac{\mathrm{d}I_j(t)}{\mathrm{d}t} = \underbrace{\lambda_j(t)\, S_j(t)}_{\text{new infections}} - \underbrace{\gamma_j\, I_j(t)}_{\text{recovery}} - \underbrace{\mu_j\, I_j(t)}_{\text{mortality}} + \underbrace{Q_j^{I,\,\mathrm{in}}(t) - Q_j^{I,\,\mathrm{out}}(t)}_{\text{movement}}. \tag{7}$$

In equation (6) the recruitment term $\mu_j N_j$ replenishes the susceptible pool at the same rate as natural mortality, maintaining a quasi-constant herd size in the absence of epidemic dynamics. Population closure is verified by adding (6) and (7):

$$\frac{\mathrm{d}N_j(t)}{\mathrm{d}t} = \left[Q_j^{S,\,\mathrm{in}}(t) + Q_j^{I,\,\mathrm{in}}(t)\right] - \left[Q_j^{S,\,\mathrm{out}}(t) + Q_j^{I,\,\mathrm{out}}(t)\right], \tag{8}$$

confirming that changes in total herd size arise only from net animal movements, as expected for a closed national herd subject to trade.

Wildlife reservoir dynamics

The reservoir $W(t)$ evolves according to:

$$\frac{\mathrm{d}W(t)}{\mathrm{d}t} = \underbrace{\xi \sum_{j=1}^{3} I_j(t)}_{\substack{\text{spillover from} \\ \text{infected cattle}}} - \underbrace{\delta\, W(t)}_{\substack{\text{badger recovery} \\ \text{or mortality}}}. \tag{9}$$

At quasi-steady state (i.e., $\mathrm{d}W/\mathrm{d}t \approx 0$), the equilibrium reservoir intensity is $W^* = (\xi/\delta) \sum_j I_j^*$, which is directly proportional to total cattle prevalence. The reservoir thus amplifies persistence in Cluster 3 herds and sustains low-level endemic infection in Cluster 2 herds even when within-herd transmission alone would not support endemicity.

Complete System Summary

Substituting equations (3)–(5) into (6)–(7) yields the fully expanded governing system for $j \in \{1, 2, 3\}$:

$$\frac{\mathrm{d}S_j(t)}{\mathrm{d}t} = \mu_j N_j - \left[\beta_j \frac{I_j}{N_j} + \alpha_j \sum_{k \neq j} w_{jk} \frac{I_k}{N_k} + \sigma_j W\right] S_j + \gamma_j I_j - \mu_j S_j + \sum_{k \neq j} m^S_{kj} S_k - \sum_{k \neq j} m^S_{jk} S_j, \tag{10}$$

$$\frac{\mathrm{d}I_j(t)}{\mathrm{d}t} = \left[\beta_j \frac{I_j}{N_j} + \alpha_j \sum_{k \neq j} w_{jk} \frac{I_k}{N_k} + \sigma_j W\right] S_j - \gamma_j I_j - \mu_j I_j + \sum_{k \neq j} m^I_{kj} I_k - \sum_{k \neq j} m^I_{jk} I_j, \tag{11}$$

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \xi\Big(I_1 + I_2 + I_3\Big) - \delta W, \tag{12}$$

where all time arguments have been suppressed for compactness.

The complete system comprises $2 \times 3 + 1 = 7$ ordinary differential equations coupled through the shared reservoir (12) and through the neighbourhood pressure terms $L_j(t)$ embedded in the force of infection.

Cluster-Specific Parameterisation and Biological Interpretation

Cluster 1 — self-limiting breakdowns. Low β_1 ensures that the basic reproduction number within the herd ($R_{0,1} = \beta_1/(\gamma_1 + \mu_1)$) remains below or marginally above unity, so that most introductions extinguish without sustained transmission. Low σ_1 further reduces the probability of wildlife-driven reintroduction. The system is attracted to the disease-free equilibrium.

Cluster 2 — recurrence-driven endemicity. Very low β_2 implies that within-herd $R_{0,2}$ is sub-critical, yet elevated σ_2 and substantial $m^I_{k\to 2}$ rates continuously seed new breakdowns. The endemic state is sustained by repeated external reintroduction rather than self-sustaining transmission, explaining why Cluster 2 shows the fastest decline in breakdown-free survival probability despite limited within-breakdown spread.

Cluster 3 — high-intensity sustained outbreaks. High β_3 drives supra-critical within-herd dynamics. Geographic clustering amplifies $L_3(t)$, creating a positive feedback loop in which dense spatial networks of Cluster 3 herds mutually elevate each other's force of infection. Prolonged γ_3^{-1} (long breakdown duration) further sustains epidemic persistence.

Parameter Estimation Strategy

Cluster-specific parameters are estimated directly from surveillance records as follows:

1. γ_j: Fit the empirical distribution of breakdown durations within each cluster; $\hat{\gamma}_j = 1/\bar{d}_j$, where $\bar{d}_j$ is the cluster mean breakdown duration.
2. β_j: Estimated from the median time-to-peak reactor count within each cluster's outbreak trajectory, or via maximum-likelihood fitting to within-breakdown incidence curves.
3. σ_j: Calibrated to the empirical proportion of breakdowns within cluster j that exhibit zero within-herd spread (i.e., a single reactor), interpreted as index cases arising from wildlife exposure rather than herd-to-herd transmission.
4. $m_{jk}^I(t)$: Derived from the cattle-movement database, stratified by source-herd breakdown status at the time of each registered movement.
5. α_j: Mapped from the odds ratios of the multinomial logistic regression — specifically, the coefficient associated with the count of New Breakdown neighbours, which directly quantifies the neighbourhood susceptibility of each cluster type.
6. ξ, δ: Estimated from published badger-demographic and spillover studies, or treated as fixed sensitivity-analysis parameters if telemetry data are unavailable.

Connection to Stochastic Simulation

The deterministic system (10)–(12) provides the *mean-field* trajectory around which the stochastic process fluctuates. Under a continuous-time Markov chain (CTMC) implementation — as in the SimInf package — each elementary event (infection, recovery, birth, death, movement) fires as an exponentially distributed waiting time whose rate equals the corresponding term in the differential equations. As the number of stochastic realisations $n \to \infty$, the ensemble mean $\mathbb{E}[\,I_j(t)\,]$ converges to the deterministic solution $I_j(t)$ by the law of large numbers for Markov chains.

The stochastic layer captures phenomena invisible to the deterministic model:

- **Stochastic extinction**: small herds (common in Cluster 1) may clear infection by chance even when $R_0 > 1$, inflating the apparent breakdown-free fraction.
- **Outbreak variability**: Cluster 3 simulations will exhibit a right-skewed distribution of outbreak sizes, consistent with observed superspreading dynamics.
- **Cluster recovery**: if the model is correctly specified, applying the same k-means or PAM clustering algorithm to simulated outbreak trajectories should recover three clusters whose centroids and membership proportions match the empirical Clusters 1–3. Systematic deviation identifies model components requiring recalibration.